

Desk controller

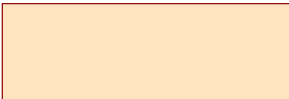
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Change history

1.0.0	23/09/2026	First release
1.1.0	18/10/2026	Migration to KiBot template
1.0.0	TBD	TBD
1.0.0	TBD	TBD
1.0.0	TBD	TBD
1.0.0	TBD	TBD
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1.0.0	TBD	TBD

System overview



File: System_overview.kicad_sch

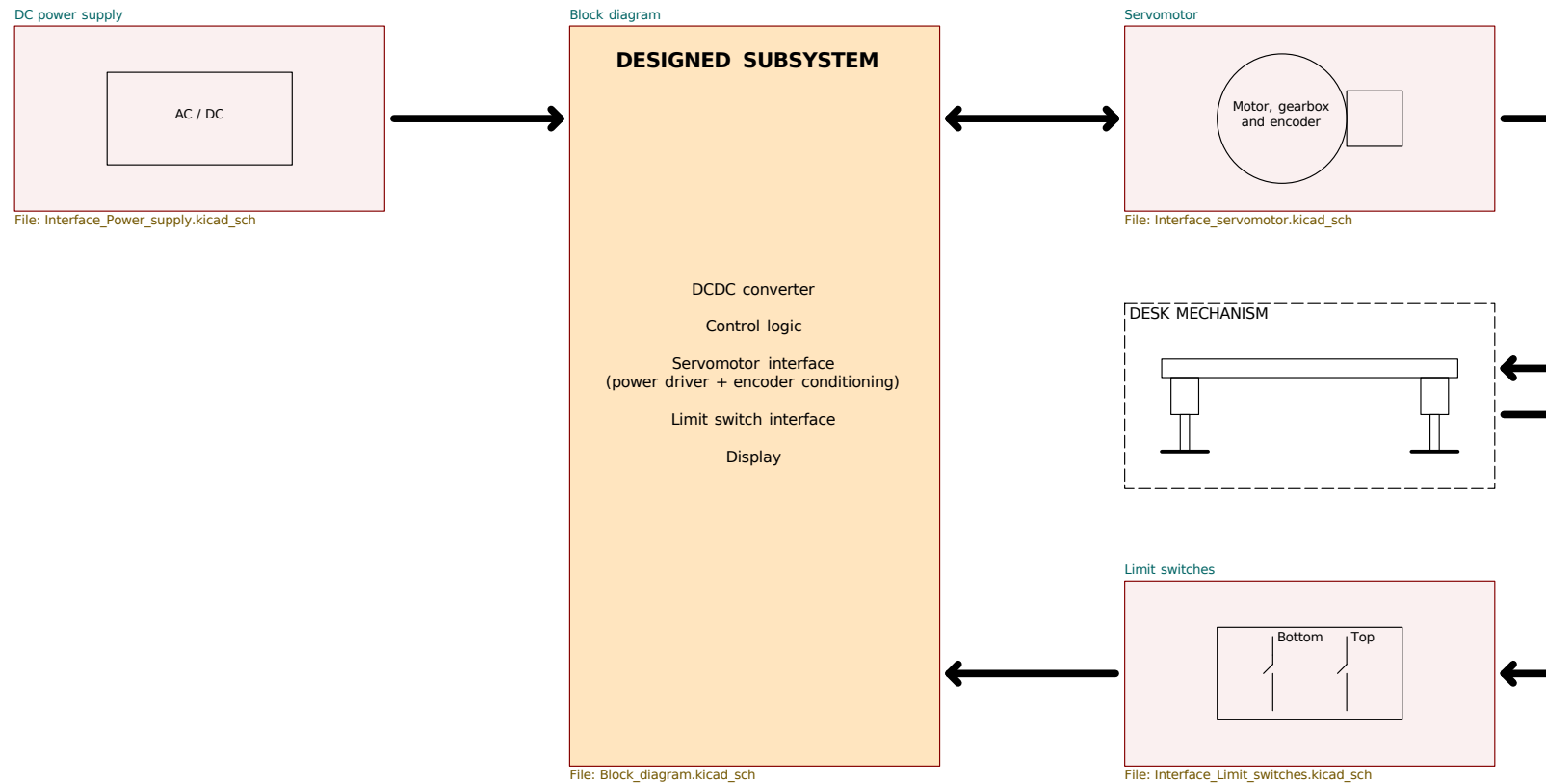


Desk controller

Notes

Sheet: Root	
File: ktest.kicad_sch	
Rev: 1.0.1	Drwn: FLR
Mat: Concept	Rev'd: FLR
Date: 2026-10-06	RLs'd: FLR
Dwg: NA	Size: A4
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System overview



This sheet shows the overall system and the interfaces to external components. External components are shown for system context only and are not part of this schematic. Only the "DESIGNED SUBSYSTEM" is covered by this schematic.

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Desk controller

Notes

Sheet: System overview

File: `System_overview.kicad_sch`

Rev: 1.0.1

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Drwn: FLR

Rev'd: FLR

RLs'd: FLR

Size: A4

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DC power supply

Interface specification

Barrel connector (Dell type, with ID pin)

Pin

- 1 Supply - 19.5 V, 3 A (inner shell)
- 2 ID (internal pin)
- 3 GND (outer shell)
- SH Shield

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Notes

Sheet: DC power supply	
File: Interface_Power_supply.kicad_sch	
Rev: 1.0.1	Drwn: FLR
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Date: 2026-10-06	RLs'd: FLR
Dwg: NA	Size: A4
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Interface specification

JST B6B-XH-A / JST XHP-6 connector

Pin	Description
1	Encoder B output (VCC)
2	Encoder A output (VCC)
3	GND
4	VCC (3.3 - 5V)
5	Motor B (3 A, 12-24 V)
6	Motor A (3 A, 12-24 V)



Desk controller	Sheet: Servomotor	
	File: Interface_servomotor.kicad_sch	
	Rev: 1.0.1	Drwn: FLR
	Mat: Concept	Rev'd: FLR
	Date: 2026-10-06	RLs'd: FLR
Notes	Dwg: NA	Size: A4
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Limit switches

Interface specification

JST B4B XH A / JST XHP-4 connector

Pin Description

- 1 Bottom limit switch (Normally open, switch closes to GND)
- 2 GND
- 3 GND
- 4 Top limit switch (Normally open, switch closes to GND)

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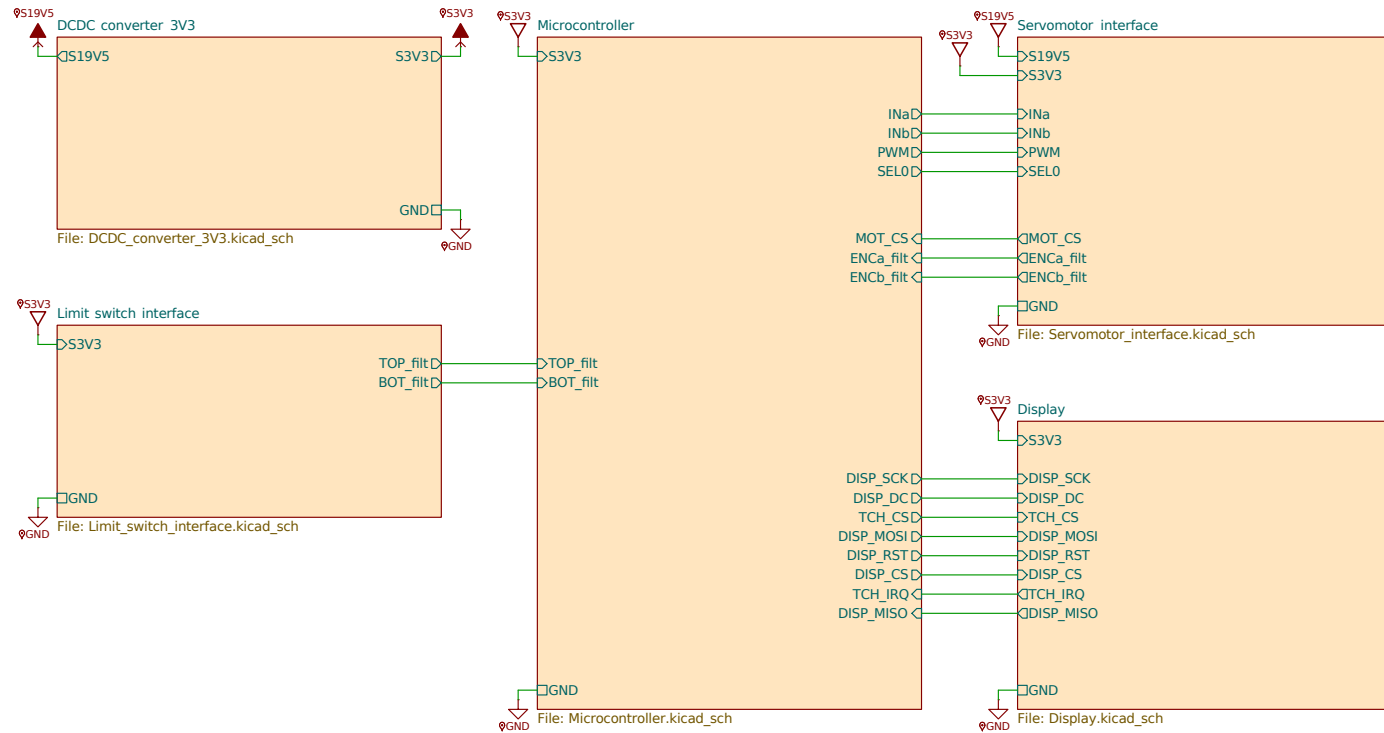


Desk controller

Notes

Sheet: Limit switches			
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Mat: Concept	Rev'd: FLR		
Date: 2026-10-06	Rls'd: FLR		
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Block diagram



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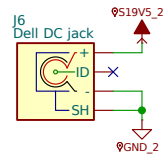
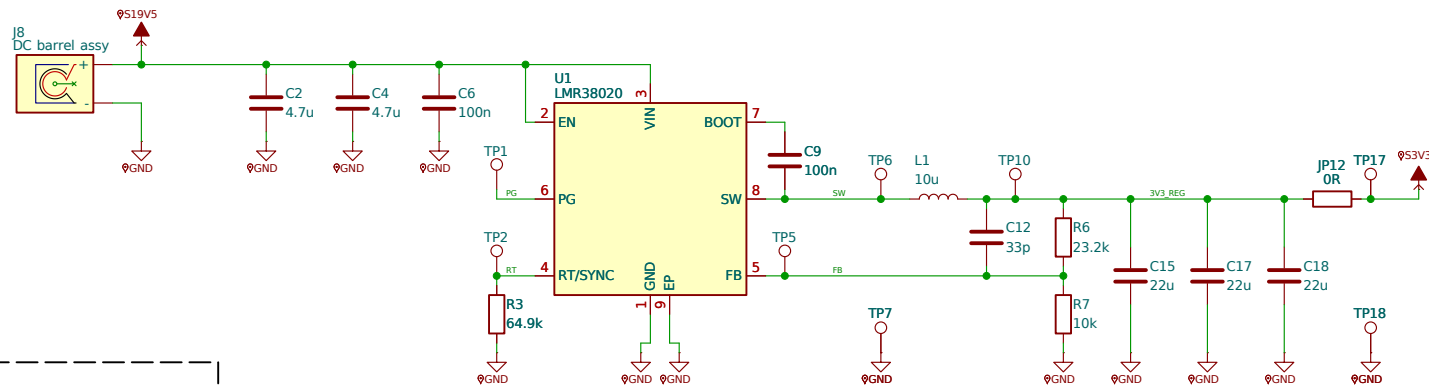
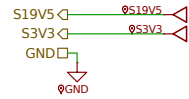
Desk controller

Notes

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Rev: 1.0.1	Drwn: FLR
Mat: Concept	Rev'd: FLR
Date: 2026-10-06	RLs'd: FLR
Dwg: NA	Size: A4
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DCDC converter 3V3

Block interface



J6 is fitted to a small "standalone" PCB which will then be soldered into the "main" PCB. This standalone PCB + connector are what make the "DC barrel assy" used in J8 position.

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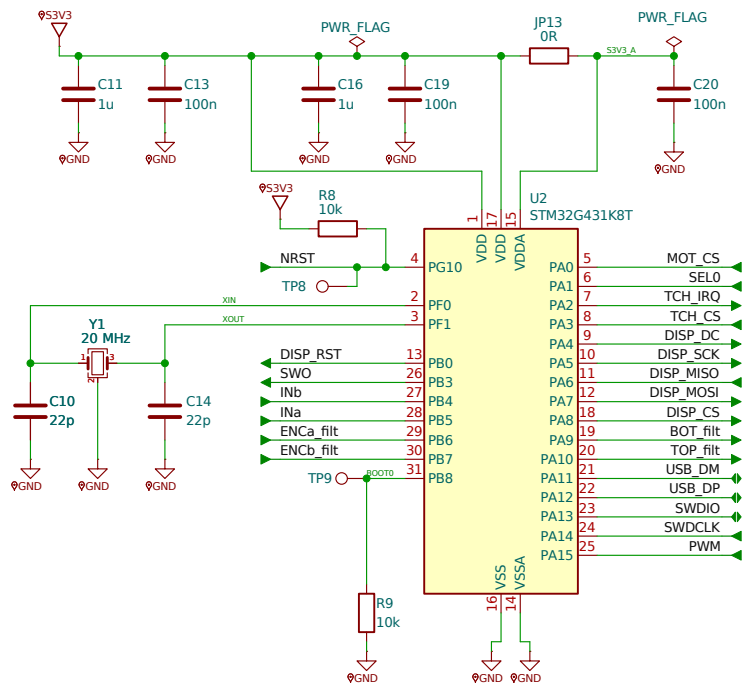


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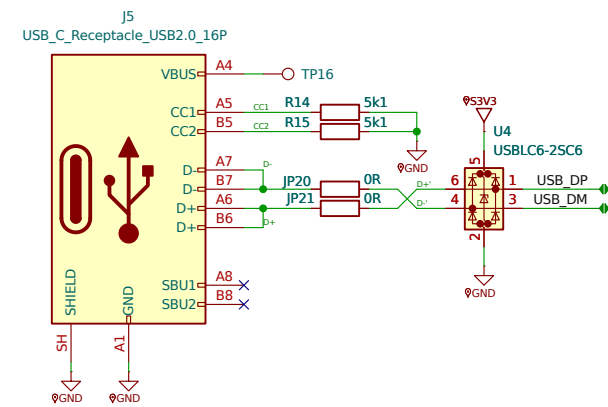
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Mat: Concept	Rev'd: FLR
Date: 2026-10-06	Rs'd: FLR
Dwg: NA	Size: A4
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Pinout diagram for the STM32F103C8T6 microcontroller. The diagram shows 20 pins on the left and right sides of the package. The left side pins are labeled: S3V3, INa, INb, PWM, SELO, MOT_CS, ENCa_filt, ENCb_filt, TOP_filt, BOT_filt, DISP_RST, TCH_CS, DISP_CS, DISP_SCK, DISP_MOSI, DISP_DC, DISP_MISO, TCH_IRQ, and GND. The right side pins are labeled: 05S3V3, INa, INb, PWM, SELO, MOT_CS, ENCa_filt, ENCb_filt, TOP_filt, BOT_filt, DISP_RST, TCH_CS, DISP_CS, DISP_SCK, DISP_MOSI, DISP_DC, DISP_MISO, TCH_IRQ, and GND. The GND pin is connected to a ground symbol.



USB interface (serial comms port)



Pin configuration diagram for the J4 connector. The diagram shows a 10-pin connector with pins 1-5 on the left and 6-10 on the right. The connections are as follows:

- Pin 1: SWO
- Pin 2: SWDCLK
- Pin 3: SWDIO
- Pin 4: SWO
- Pin 5: NRST
- Pin 6: SWO
- Pin 7: SWDIO
- Pin 8: SWDCLK
- Pin 9: SWO
- Pin 10: SWDIO

The diagram also shows connections to GND, 53V3, and a 14-pin connector.



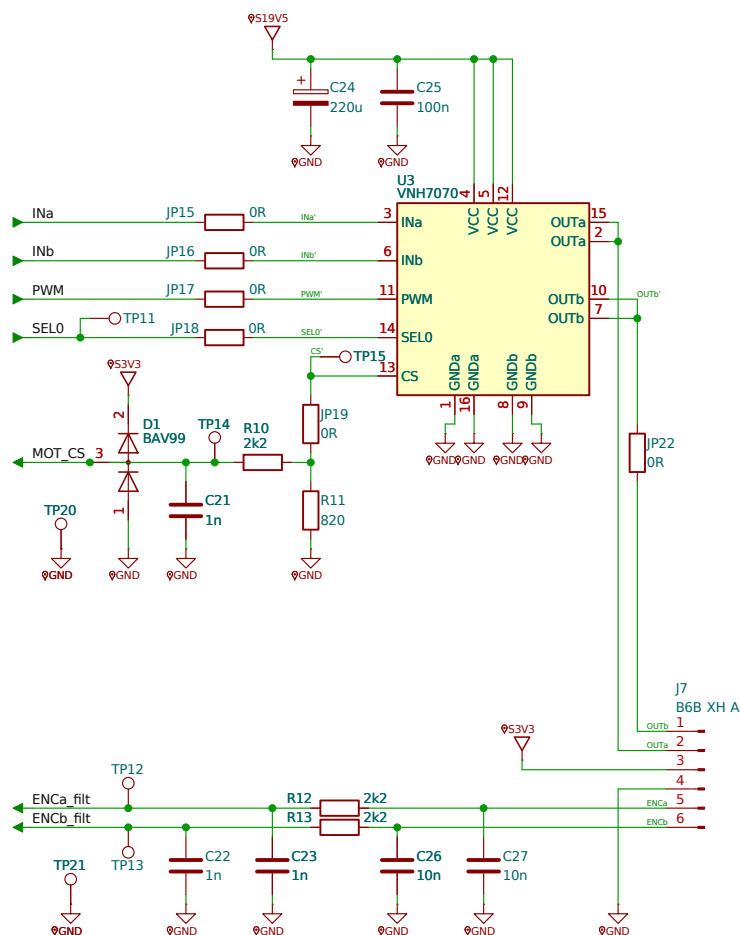
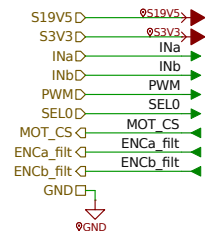
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Servomotor interface

Block interface



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Desk controller

Notes

Sheet: Servomotor interface

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Mat: Concept

Date: 2026-10-06

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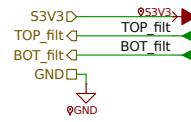
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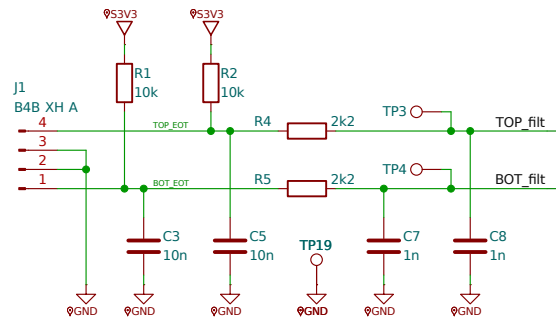
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Block interface



Limit switch interface



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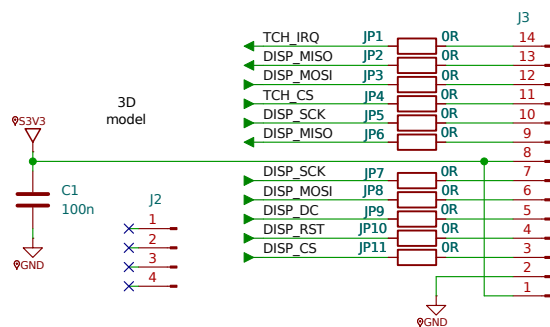
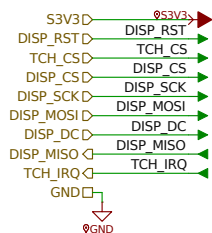


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Block interface



TFT1
TFT_2.4in_320x240

The display will not be directly soldered to the PCB. Instead, generic 2.54 mm pitch sockets J2 and J3 will be used as connectors.

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Desk controller

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